

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, SPRING SEMESTER 2021-2022

Mathematics for Computer Scientists 2

Time allowed: Two Hours

Answer ALL FIVE Questions

INSTRUCTION FOR STUDENTS:

- The university's honour code applies to this individual short time assessment. The submitted answers **must be your own work** where **answers are attempted without any help from others**. University academic misconduct rules apply here in the case of plagiarism, collusion, or false authorship.
- Write the name and student ID at the top of your answer sheets.
- Students are not allowed to ask questions during the short time assessment in accordance with the no shout-out policy.
- You may answer the questions in any order but write the question numbers correctly and clearly.
- At the end of the exam, **upload your** solutions and name the file as **StudentID.pdf** Upload your solution file onto **Moodle**.
- Please ensure that you upload the correct files to the folder labeled as "Final examination submission folder" in Moodle. Request for resubmission will not be accepted after the submission.
- Students should stop writing the answers at the end of exam time. Answer scripts must be submitted to Moodle within **thirty minutes** after the end of the exam. In case of any technical issues, the students need to contact the module convener in this period.

IMPORTANT: Late submission will not be accepted.

QUESTION 1: Systems of linear equations- Matrix Theory

Let us consider the following given system of linear equations

$$\begin{cases} 2x - y + z = 1 \\ 4x - 2y + 2z = 2 \\ 6x - 3y + 3z = 3 \end{cases}$$

- a) Find the rank of the incomplete matrix **A** and complete matrix **A^c** associated with the system. On the basis of the calculated ranks, assess whether or not the system is compatible.

[05 Marks]

- b) Calculate, if possible, the solution or the solutions satisfying the system of linear equations. If the system allows multiple solutions calculate the number of solutions satisfying the system.

[05 Marks]

- c) If the known terms are changed to zero, then the system of linear equations is

$$\begin{cases} 2x - y + z = 0 \\ 4x - 2y + 2z = 0 \\ 6x - 3y + 3z = 0 \end{cases}$$

Calculate the rank of the complete matrix and calculate, if possible, the solution or the solutions satisfying the system of linear equations. If the system allows multiple solutions calculate the number of solutions satisfying the system.

[05 Marks]

- d) Lets the known terms be changed and the system of linear equations is

$$\begin{cases} 2x - y + z = 1 \\ 4x - 2y + 2z = 2 \\ 6x - 3y + 3z = 2 \end{cases}$$

Find the rank of the complete matrix and calculate, if possible, the solution or the solutions satisfying the system of linear equations. If the system allows multiple solutions, find the number of solutions satisfying the system.

[05 marks]

NOTE: For the above question parts (a, b, c, and d) show and comment on, all the steps in answers.

End of Question 1: Total 20 marks

QUESTION 2: Systems of linear equations- Cramer's Method/Direct Methods

a) Let us consider the following given system of linear equations

$$\begin{cases} 2x - y + 2z = 1 \\ x + 2z = 1 \\ 2x - y = 1 \end{cases}$$

Determine whether or not this is a Cramer's system. If this is a Cramer's system, solve it (find $x, y, \text{ and } z$) by applying Cramer's method.

[06 Marks]

b) Let us consider the following given system of linear equations

$$\begin{cases} x + 2y - 3z = 1 \\ 2x + y - z = 1 \\ x - 4y + 4z = 1 \end{cases}$$

Apply the method of Gaussian elimination to determine the equivalent triangular system. When the triangular system has been obtained then solve the system (find $x, y, \text{ and } z$).

[06 Marks]

c) Let us consider the following given system of linear equations

$$\begin{cases} x + 2y + 2z = 1 \\ 2x + 5y + z = 2 \\ 2x + 8y + z = 5 \end{cases}$$

Perform the LU factorisation of the incomplete matrix associated with the system of linear equations. After the factorisation, solve the original system (find $x, y, \text{ and } z$) by making use of the LU factorisation.

[08 Marks]

NOTE: For the above question parts (a, b, and c) show and comment on, all the steps in answers.

End of Question 2: Total 20 marks

QUESTION 3: Linear Mapping

Let us consider the following linear mapping $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ in the variables $x = (x_1, x_2, x_3)$

$$f: (x_1 + x_2 + x_3, x_1 - x_3, x_2 + x_3)$$

a) Determine the image of the mapping $Im(f)$ (that is find a basis spanning $Im(f)$).

[05 Marks]

b) Determine the kernel of the mapping $ker(f)$ (that is find a basis spanning $ker(f)$).

[05 Marks]

c) Determine the mapping f is injective, surjective, or bijective.

[05 Marks]

d) Determine the dimension of kernel $ker(f)$ and image $Im(f)$, and then verify the validity of the Rank-Nullity Theorem.

[05 Marks]

NOTE: For the above question parts (a, b, c and d) show and comment on, all the steps in answers.

End of Question 3: Total 20 marks

QUESTION 1: Matrices/System of Linear Equations

- a) Determine for what values of the parameter λ , the system is determined and incompatible.

$$\begin{cases} 4x - y = 1 \\ \lambda x - \lambda y = 1 \\ (\lambda + 2)x + (\lambda - 1)y - z = -2 \end{cases}$$

[10 Marks]

- b) Let us consider a matrix $\mathbf{A} \in \mathbb{R}_{n,n}$. Prove that the rows of matrix $\mathbf{A}$ are linearly dependent if and only if the columns are linearly dependent.

[10 marks]

Note: To perform the proof, use relevant definitions, theorems, lemmas, corollaries, and propositions studied in this module.

End of Question 1: Total 20 marks

QUESTION 2: Vector Spaces/Linear Mapping

- a) Let us consider the following three vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 4 \\ 4 \\ 3 \end{pmatrix}$$

Determine whether or not the vectors are linearly dependent. Find the linear span, and identify (if possible) one or more vectors that can be removed from the span without changing the spanning set.

[07 Marks]

- b) Let us consider a mapping $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ whose associated matrix $\mathbf{A}$ has rank $\rho \mathbf{A} = 1$. If $\mathbf{v}, \mathbf{u}$, and $\mathbf{w}$ are three linearly independent vectors then determine whether or not $f(\mathbf{v}), f(\mathbf{u})$ and $f(\mathbf{w})$ are linearly dependent and proportional? Justify your answer.

[06 Marks]

- c) Let consider the following vector spaces $(U, +, \cdot)$ and $(V, +, \cdot)$ where

$$U = \{(x, y, z) \in \mathbb{R}^3 \mid \{x - y + z = 0\}\}$$

$$V = \{(x, y, z) \in \mathbb{R}^3 \mid \{x - 2y - 2z = 0\}\}$$

- c1. Find the basis and dimension of $U, V, U \cap V$ and $U + V$.

[05 Marks]

- c2. What is the geometrical interpretation of $U \cap V$ and $U + V$?

[02 Marks]

Note: To perform the proof in parts a) and b), use relevant definitions, theorems, lemmas, corollaries, and propositions studied in this module.

End of Question 2: Total 20 marks